INTERNATIONAL MONETARY FUND

MALDIVES

July 29, 2021

STAFF REPORT FOR THE 2021 ARTICLE IV CONSULTATION—DEBT SUSTAINABILITY ANALYSIS

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Prepared by the staffs of the International Monetary Fund and the International Development Association.

Maldives **Joint Bank-Fund Debt Sustainability Analysis**	
Risk of external debt distress:	High
Overall risk of debt distress:	High
Granularity in the risk rating:	Not sustainable under baseline
Application of judgement:	No

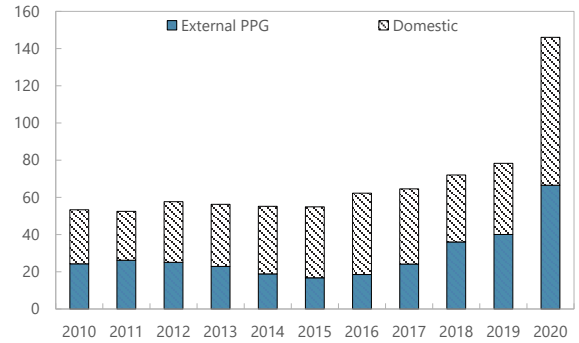
The Maldives continues to be at a high risk of debt distress. Since the April 2020 Debt Sustainability Analysis (DSA), all end-2020 debt indicators have deteriorated, mostly reflecting the larger than expected impact of COVID-19 on the economy, especially tourism related sectors.[1] Projections for the debt level and fiscal deficit have worsened, driven by both a lower projected GDP path and the authorities' expanded medium-term public investment program. The April 2021 Sukuk issuance rolled over 77 percent of the 2020 Eurobond, decreasing the associated rollover risk. The authorities are, and have been, current on debt service obligations to both domestic and external creditors. Nonetheless, unlike the April 2020 DSA, protracted breaches in several debt indicators over the medium term make the assessment of debt unsustainable under the baseline projections. Securing debt sustainability requires balanced and sustained fiscal consolidation, conservative debt management, and continued strong growth. With prominent downside risks to the outlook, debt dynamics will remain vulnerable to adverse shocks to growth, interest rates, and the fiscal position.

[1] The debt carrying capacity of the Maldives is classified as weak. The composite indicator is estimated at 2.53 and it is based on the April 2021 WEO and 2019 World Bank's Country Policy and Institutional Assessment (CPIA) score.

BACKGROUND AND DEVELOPMENTS ON DEBT

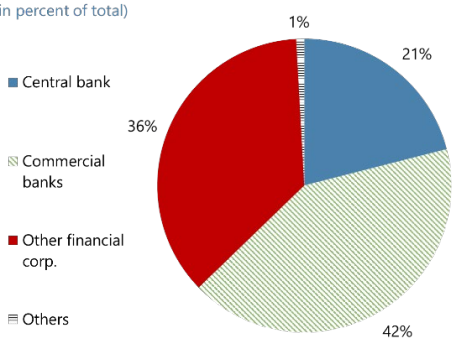
1. The total PPG debt-to-GDP ratio increased from 78 percent in 2019 to 146 percent in 2020, reflecting both the contraction in nominal GDP by about 34 percent and an expansion in PPG nominal debt by 24 percent. Total Public and Publicly Guaranteed (PPG) debt stood at USD 5,459 million, or 146 percent of GDP at end-2020. This implies a nominal increase of about USD 1048 million (24 percent growth) compared to end-2019 debt levels. Both external and domestic debt increased during 2020.[2]

Total Public and Publicly Guaranteed Debt
(in percent of GDP)



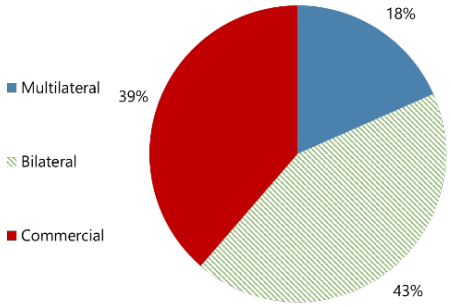
Sources: Maldives Ministry of Finance and IMF staff projections

Claims on the Central Government by Ownership, 2020
(in percent of total)



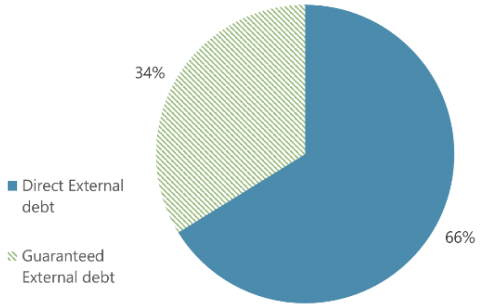
Source: Maldives Monetary Authority

External PPG Debt by Creditor Type, 2020
(in percent of total)



Source: Maldives Ministry of Finance

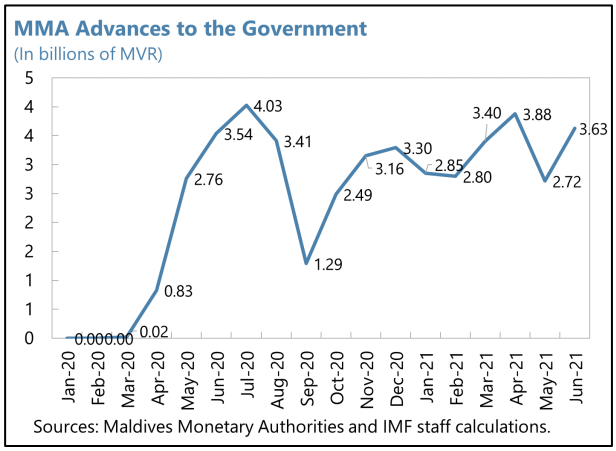
External PPG Debt, 2020
(in percent of total)



Source: Maldives Ministry of Finance

[2] The Maldives availed of swap arrangements with the Reserve Bank of India (RBI) for USD 400 million by the end of 2020. These amounts were availed in two tranches, USD 150 million in April and USD 250 million in December. The USD 150 million was repaid in January 2021. The swap arrangement with the RBI is for the purpose of reserve management and there is no indication it is being used for fiscal purposes. Thus, the swap with RBI is excluded from public debt.

2. Domestic PPG debt stood at USD 2,972 million or 79.5 percent of GDP by end 2020. This is an increase of around USD 817 million from 2019 and includes USD 214 million in advances from the Maldives Monetary Authority (MMA) to the central government and a USD 250 million bond purchased by the State Bank of India Malé Branch as budget support to buffer the impact of COVID-19. Outstanding public debt of the central government is owed mostly to commercial banks at 42 percent,[3] followed by other financial corporations, including pension funds at 36 percent, and the central bank at 21 percent. Domestic debt outstanding to the central bank includes around USD 390 million (or 13 percent of domestic PPG debt) from a bond issued in 2014 to convert debt held by the MMA in the ways and means account. To bridge the financing of the government during the COVID-19 crisis, in end-April 2020 the parliament of the Maldives approved the temporary suspension of elements of the Fiscal Responsibility Act (FRA) to allow for a one-year expansion of the cap on government advances with the MMA to MVR 4.4 billion (around USD 286 million). The Ministry of Finance has requested from parliament a one-year extension to the increase in the cap on government advances to help bridge financing needs in 2021. Total government advances with the MMA stood at around USD 214 million by end-2020.



Sources: Maldives Monetary Authorities and IMF staff calculations.

3. External PPG debt stood at USD 2,488 million, or 66.5 percent of GDP on end-2020. Around 34 percent of external PPG debt (USD 841 million) is guaranteed by the government and 67 percent of it is associated with the financing of large housing projects by the state-owned enterprise (SOE) the Housing Development Corporation (HDC) . Guaranteed external debt is owed mainly to commercial creditors (around 72 percent) with the rest bilateral (26 percent) and a small amount to a multilateral creditor (2 percent). Direct external debt is largely held by bilateral creditors (52 percent), with multilateral creditors holding 27 percent and commercial creditors holding 21 percent. Chinese creditors hold around 53 percent of External PPG debt. During 2020, the Maldivian government received exceptional external COVID-19 funding amounting to around USD 196 million in loans, USD 28.9 million from the IMF's Rapid Credit Facility (RCF) approved on April 22, 2020, and USD 56 million in grants.[4] The Maldives has issued two sovereign Eurobonds with a total face value of USD 350 million, and in March 2021 issued a Sukuk bond with a face value of USD 200 million that will be used to rollover 77 percent of the USD 250 million maiden Eurobond to April 2026.[5] The Sukuk subscription was reopened on April 27 under the same conditions, and

[3] The SBI loan is included in commercial bank claims on the central government.

[4] Around 90 percent of grants were provided by the Asian Development Bank (USD 27.3 million), World Bank (USD 16.1 million), and Japan (USD 7.5 million).

[5] The Sukuk has a coupon of 9.875 percent, yield of 10.5 percent at issuance, maturity of five years (in 2026), and a bullet repayment. The maiden Eurobond was issued in 2017, has a face value of USD 250 million, coupon of 7 percent, maturity of five years (in 2022), and a bullet repayment. The privately placed Eurobond was issued in 2018, has a face value of USD 100 million, coupon of 5.5 percent, original maturity of five years (in 2023), extended by three years (to 2026), and a bullet repayment.

the government placed an additional USD 100 million.[6] The Maldives applied for the G20 Debt Service Suspension Initiative (DSSI) and received relief from most bilateral official creditors totaling around USD 19 million in 2020. They also applied for both extensions to the DSSI from creditors that provided suspension on the original request. Benefits from the extension are estimated at around USD 69 million in 2021.[7]

4. This debt sustainability analysis includes PPG external and domestic debt. Public debt includes the debt of the central government, including guarantees to SOEs. Other components do not apply to the Maldives. External debt is defined by the residency criteria. The government domestically issues some USD denominated debt (outstanding as of December 2020 was USD 205 million) and has recently borrowed USD 250 million from the State Bank of India branch in Malé. These amounts are classified as domestic debt since the holders are residents. In keeping with the government's commitment to transparency in debt data, information about guarantees to SOEs and other details of the public debt are publicly disclosed on the webpage of the Ministry of Finance.

Public Debt Coverage

Subsectors of the public sector	Sub-sectors covered
Central government	X
State and local government	
Other elements in the general government	
o/w: Social security fund	
o/w: Extra budgetary funds (EBFs)	
Guarantees (to other entities in the public and private sector, including to SOEs)	X
Central bank (borrowed on behalf of the government)	
Non-guaranteed SOE debt	

[6] As an IDA-eligible country at high risk of external debt distress, Maldives is subject to a zero ceiling on external non-concessional borrowing under IDA's Sustainable Development Finance Policy (SDFP).

[7] The DSSI is a G20 initiative to help IDA-eligible countries cover the health costs of COVID-19. The original DSSI consisted of a suspension of interest and principal repayments due between May 1 and Dec. 31, 2020 from official bilateral creditors on direct debt. Private creditors were encouraged to participate. Two extensions of the DSSI were applied to suspend interest and principal payments falling in 2021. The payment of suspended amounts will be NPV neutral. The suspended amounts from the original DSSI are to be repaid in three years with one-year grace. The amounts from the extension are to be repaid over 5 years with one-year grace.

BACKGROUND ON MACRO FORECASTS

5. The impact of COVID-19 on the Maldives has been worse than projected at the time of the last DSA which was prepared for the 2020 RCF report.

- **Growth and inflation.** The tourism sector makes up around one quarter of value-added GDP and is a major driver of growth. The baseline of the 2020 RCF report, following global assumptions for the April 2020 WEO, assumed a shorter-lived contraction in economic activity, with the tourism sector recovering to pre-COVID levels in Q4 2020. In contrast, the depth of the recession was much larger with growth contracting by 32 percent in 2020 and with tourism in December 2020 recovering to around 56 percent of pre-COVID levels. The pace of the recovery in tourism since December 2020 in the Maldives is promising and ahead of its competitors. Tourist arrivals in 2021 to date, June 30, continue to be at around 59 percent of their pre-COVID levels despite no arrivals from China, the largest market for tourists in 2019. Most tourist arrivals post-COVID have been from India (17 percent), Russia (23 percent), and other European countries (19 percent). Between the temporary halt in arrivals for tourists originating in South Asia on May 13 and end-June, 2021, tourist arrivals have been around 52 percent of their pre-COVID levels.[8] A partial GDP rebound is projected for 2021 with arrivals projected to be around 60 percent of the 2019 pre-COVID level. Medium-term growth projections are supported by a gradual return to the level of tourist arrivals projected for 2020 pre-COVID-19 by 2023 and the expansion of the airport to meet the pre-COVID-19 higher demand from airlines. Prices contracted in 2020 by about 1.6 percent and the outlook on inflation remains moderate.

- **Primary fiscal deficit.** The approved 2021 budget and 2022-23 medium-term projections envision an expansion of capital spending. The framework incorporates the revised capital expenditure plans of the authorities amounting to 16 percent of GDP in 2021, and 14 and 12 percent of GDP in 2022 and 2023, respectively. This is compared to capital expenditure of 9 percent of GDP in 2019 and an average of 8 percent of GDP during 2010-18. As such, the medium-term capital spending plans of the authorities imply that the primary fiscal deficit will remain elevated at around 14 percent of GDP in 2021 and decline to around 6 percent of GDP in 2023.

- **Domestic financing.** Assumptions on domestic financing are broadly in line with those of the 2020 RCF DSA. As of June 2020, around 62 percent of domestic financing in the Maldives was short term (maturity of less than one year). Interest rates on short-term debt have ranged between 3.7 and 4.5 percent since December 2016. Over the medium term, we assume around 60 percent of domestic financing is short term with an interest rate of 4 percent and that the share of long-term debt gradually increases over time. Unsecured financing is assumed to be met from domestic creditors. Domestic banks and pension funds are projected to remain the main holders of government debt.

[8] The halt was lifted for travelers originating in South Asia on July 15, 2021.

Change in Macro Assumptions

	2020	2021	2022	2023	2024	2025	2026	27-40
Real GDP growth (in percent change)								
Current	-32.0	18.9	13.2	12.1	6.1	5.4	5.4	4.6
Previous	-8.1	13.2	6.4	5.9	5.4	5.4	5.4	4.6
Inflation								
Current	-1.6	1.4	2.3	1.9	1.8	2.0	2.0	2.0
Previous	1.5	1.5	2.0	2.0	2.0	2.0	2.0	2.0
Primary fiscal deficit (in percent of GDP)								
Current	20.0	14.2	9.0	5.7	3.9	3.3	4.0	0.5
Previous	10.1	4.4	3.1	1.9	1.4	1.1	0.9	0.3
Non-interest current account deficit (in percent of GDP)								
Current	27.0	13.7	11.9	7.1	7.2	6.9	6.6	4.6
Previous	22.2	10.5	7.9	7.8	8.0	7.8	6.7	4.7
Grants (in percent of GDP)								
Current	2.5	2.1	2.7	2.3	1.4	0.9	0.8	0.3
Previous	2.6	1.4	1.0	0.7	0.6	0.6	0.5	0.3
Capital Expenditures (in percent of GDP)								
Current	14.6	15.6	13.9	12.4	11.3	10.2	10.8	7.0
Previous	8.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Public Debt (in percent of GDP)								
Current	146.0	137.2	133.3	128.2	128.4	126.8	122.9	100.3
Previous	93.9	87.3	85.4	83.0	80.7	78.1	75.5	61.4

- **Non-interest current account deficit (CAD).** Driven by a decline in imports, the current account deficit contracted by around 25 percent in 2020 (or USD 371 million) though the ratio to GDP remained elevated at 30 percent. Tourism-related receipts contracted by around 56 percent in 2020 but they are projected to increase into the medium-term with the steady increase of tourist arrivals. In 2020 tourist arrivals were around 555 thousand compared to 1.7 million in 2019. Arrivals are projected to be 1 million in 2021 and gradually increase to 1.96 million by 2023, around the level projected for 2020 before the COVID-19 shock. In line with the capital expenditure plans of the authorities, capital imports are projected to be elevated in the next few years. A large part of the current infrastructure investment push by the government is financed through external loans from India Exim Bank.

6. The realism tool for the baseline projections pick up the substantial size of the authorities' capital investment plans over the next few years (Figure 3). The improvement in the primary balance-to-GDP ratio between 2020 and 2023 is driven by a return to more moderate levels of capital spending and a recovery in GDP. Given the extraordinary nature behind the swings in growth rates, largely driven by changes in tourism, the multiplier approach is not a suitable way to calculate alternative growth paths. The Government of Maldives signed lines of credit for more than USD 1.2 billion with EXIM Bank India. The largest, for USD 800 million and USD 400 million, were signed in March 2019 and October 2020 respectively to finance several infrastructure projects including the development of the Gulhifalhu Port and relocation of Male Commercial Harbour, the Greater Male' Connectivity Project, construction of social housing units, and water and sanitation projects. The growth impact may be limited due to the high import content of such projects and the primarily social focus of several projects (i.e. housing, water and

sanitation). For this reason, the contribution of government capital expenditure to growth is expected to remain close to those in the previous DSA.

7. Despite higher fiscal deficits, debt dynamics in the Maldives will be mostly driven by growth in the medium-term (Figure 4). In the past, the current account deficit and the primary deficit have been the main contributors to the accumulation of external and domestic debt, respectively. Over the next five years, growth will be the dominating driver of both external and domestic debt dynamics countering the expansion of the fiscal deficit and leading to a moderate decline in projected debt-to-GDP ratios. In contrast to the last DSA, gross financing needs now breach their indicative benchmark, reflecting both lower GDP and higher borrowing (Figure 5).

COUNTRY CLASSIFICATION AND STRESS TESTS

8. The debt carrying capacity of the Maldives remains weak. The composite indicator (CI) score is calculated at 2.53 using the April 2021 WEO and the 2019 World Bank Country Policy and Institutional Assessment (CPIA). The CI is based on a weighted average of several factors such as the country's real GDP growth, remittances, international reserves, world growth, and the CPIA score. The calculation of the CI is based on 10-year averages of the variables, across five years of historical data and five years of projections, and the corresponding CPIA.

9. Tailored stress tests for natural disasters, contingent liabilities, and market financing are relevant for the Maldives. The Maldives is susceptible to rising sea levels and has issued two sovereign Eurobonds (amounting to USD 350 million) and a Sukuk for USD 300 million. Risks from the non-guaranteed SOE debt are covered by the contingent liability shock.[9] According to the 2020 Fiscal Transparency Evaluation (FTE), while there is no central information on the volume of the size of PPPs, it is understood that there are no major PPP projects in operation. The tailored stress tests were kept to their default calibrations as these are appropriate for the Maldives. The tailored stress tests are not the most extreme shocks for any of the debt indicators. The most extreme shocks are the combination shock and the shock to exports for the PPG external debt indicators, and the shock to growth for the public debt indicators.

[9] Most SOE debt is guaranteed, hence the default calibrated shock at 2 percent of GDP is reasonable to cover the risk from non-guaranteed SOE debt.

Capacity Indicator and Applicable Thresholds

Components	Coefficients (A)	10-year average values (B)	CI Score components (A*B) = (C)	Contribution of components
CPIA	0.385	3.249	1.25	49%
Real growth rate (in percent)	2.719	7.543	0.21	8%
Import coverage of reserves (in percent)	4.052	20.175	0.82	32%
Import coverage of reserves^2 (in percent)	-3.990	4.070	-0.16	-6%
Remittances (in percent)	2.022	0.000	0.00	0%
World economic growth (in percent)	13.520	3.078	0.42	16%
CI Score			2.53	100%
CI rating			Weak	

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Weak	Weak 2.53	Weak 2.41	Weak 2.47

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	140
GDP	30
Debt service in % of	
Exports	10
Revenue	14

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	35

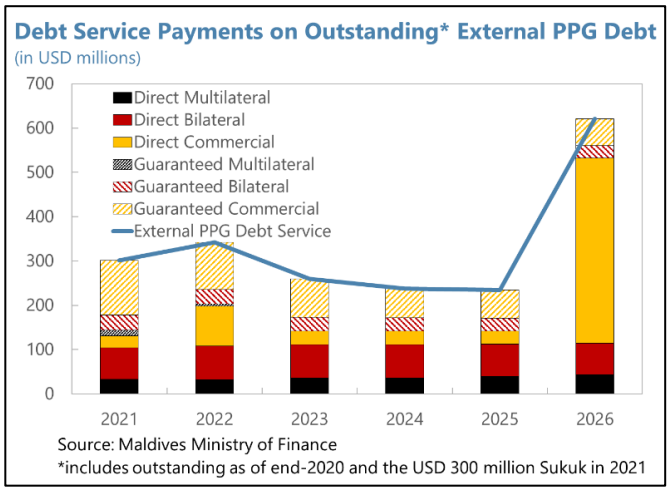
Stress Tests

		Default	Used for the analysis	Reasons for deviations from the default settings
1	The country's coverage of public debt	The central government, government-guaranteed debt		
2	Other elements of the general government not captured in 1.	0 percent of GDP	0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2	
4	PPP	35 percent of PPP stock	0.00	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5	
	Total (2+3+4+5) (in percent of GDP)		7.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

10. External debt sustainability. Maldives remains at a high risk of external debt distress. Since the last DSA, all external debt indicators have deteriorated. This is due both to the stronger contraction in growth than projected for 2020 and the higher debt-financed capital expenditures over the medium-term.[10] The PPG external public debt-to-GDP ratio in the baseline scenario is projected to decrease by around 7 percentage points to 60 percent of GDP in 2021. The PV of the PPG external debt-to-GDP ratio will remain above the threshold of 30 percent until 2031. Meanwhile, the PV of the debt-to-exports ratio is below its threshold in the baseline scenario. Both liquidity indicators, the debt service-to-exports ratio and the debt service-to-revenue ratio, breach their thresholds under the baseline scenario due to elevated debt service obligations on existing external PPG debt.



11. The authorities have been proactive in managing external rollover risks. In 2021 they issued USD 300 million in a Sukuk with a five-year maturity and a bullet repayment. However, the Sukuk was costly with a coupon of 9.875 percent and a yield of 10.5 percent. USD 200 million of the Sukuk issuance was used to repay 77 percent of the maiden USD 250 million Eurobond due in June 2022, mitigating rollover risks. The maturity of the privately placed USD 100 million bond was renegotiated in 2019 and extended by three years to 2026. Since the privately placed Eurobonds and the Sukuk will be due in 2026, this year exhibits a spike in debt service payments. The reprofiling of the debt under the DSSI has also increased debt repayment in the medium term, though amounts are relatively small.[11] Disbursements from the LOC with Exim India that occur in 2021 will start to amortize in 2026. The authorities created a sovereign development fund (SDF) to mitigate the rollover risk from these Eurobonds. Though the SDF was track path to cover the costs associated with the repayment of the Eurobonds, the COVID-19 crisis necessitated the use of most of the FX accumulated balances for MMA reserves and/or to satisfy FX demand. In the medium-term, rollover risks associated with commercial bonds, higher debt servicing costs, and continued borrowing to finance large fiscal deficits lead to protracted breaches in several external debt indicators, making debt unsustainable.

12. Overall public debt sustainability. The total PPG debt-to-GDP ratio is projected to decline by around 9 percentage points in 2021 to 137 percent on the back of higher GDP, and to gradually decline to 123 percent in 2026. Given the weak debt carrying capacity, the threshold on overall PV of debt-to-GDP is

[10] Annual IDA disbursements from 2022 until the end of the projection period are assumed on 100 percent grant terms.

[11] The authorities applied for the DSSI and both extensions. Relief in 2020 is USD 19 million, and for 2021 estimated at USD 69 million.

35 percent. The Maldives breaches this threshold for the entire projection period and remains at a high risk of overall debt distress. The domestic debt market is an important source of financing for the government with domestic debt making up 59 percent of total PPG debt on average between 2010 and 2019. It is worth noting in this context that although the Maldives qualifies for PRGT resources and thus uses the LIC DSF, the structure of the economy is closer to an EM given the level of the country's development and GDP-per-capita (above USD 10 thousand in 2019). Nonetheless, the debt-to-GDP ratio of the Maldives is above 70 percent, which according to the MAC DSA is the ratio that best predicts the occurrence of a debt distress event in EM countries. The IMF and the WB are both providing debt management technical assistance to the authorities.[12] The development of the domestic debt market to lengthen the maturity of domestic debt, especially in the case of foreign exchange denominated instruments issued in domestic markets, was recommended to the authorities. The Ministry of Finance is committed to transparency and formulates and publishes its medium-term debt strategy on its website annually. In addition, a public debt bulletin is published biannually and figures on debt outstanding are published quarterly.

RISK RATING AND VULNERABILITIES

13. The Maldives has a high risk of external debt distress and a high overall risk of debt distress. The external and fiscal positions have weakened relative to the 2020 RCF DSA due to the deeper and longer than expected COVID-19 shock as well as the authorities' expanded medium-term public investment program. Financing the larger fiscal deficit has turned more difficult and expensive, triggering the need to extend the suspension of elements of the FRA to allow the increased cap on MMA advances for one year beyond what was envisaged in April 2020. The authorities are, and have been, current on debt service obligations to both domestic and external creditors. While the recent Sukuk issuance partially mitigates rollover risks, high debt servicing costs through the medium-term and continued borrowing to finance large fiscal deficits lead to protracted breaches in several debt indicators over the medium term. As such, debt is assessed to be unsustainable under the baseline projections. Securing debt sustainability requires balanced and sustained fiscal consolidation, conservative debt management, and continued strong growth. Fiscal consolidation should include rationalizing capital spending, especially in large public investment projects carried out by the central government or SOEs, and over the medium term, mobilizing further revenues by also diversifying the tax base toward domestic resources. With prominent downside risks to the outlook, debt dynamics will remain vulnerable to adverse shocks to growth, interest rates, and the fiscal position.

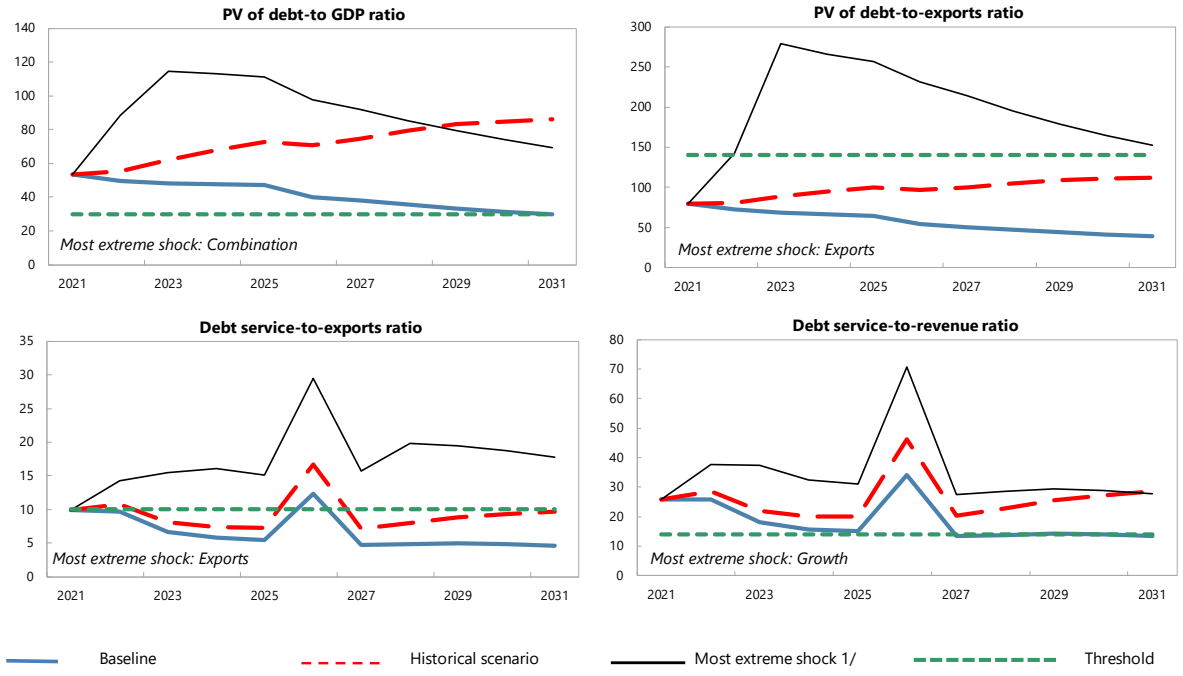
AUTHORITIES VIEWS

14. The authorities emphasized that understanding the context of the COVID-19 shock is important in making an assessment on debt sustainability. The shock to GDP was severe and caused an upward revision to all ratios to GDP. They note that the Sukuk issuance was oversubscribed signaling healthy demand for the sovereign debt of Maldives in international markets. The authorities also pointed

[12] Performance and policy actions under IDA's SDFP are being carried out to strengthen debt transparency and debt management.

out that Sukuk bonds were trading at a premium. The authorities plan to remain active in international markets and plan additional Sukuk issuances for the current year and through 2023. The authorities emphasized as an additional upside their role in proactively managing rollover risks from their outstanding Eurobonds, in 2019 by extending the maturity of the privately placed USD 100 million Eurobond by three years, and in 2021 by rolling over 77 percent of the maiden Eurobond. The authorities also highlighted that the external loans (excluding sovereign bonds) of the central government were highly concessional and have a weighted average interest rate of 2.2 percent. The weighted average interest rate for the stock of public external debt, including sovereign bonds, was around 3.5 percent. The authorities highlighted repayment capacity, as an important dimension of debt sustainability, where the deterioration was less severe and is expected to recover faster than stock ratios of GDP. The authorities attribute the ongoing capital spending expansion to a large extent to the infrastructure projects initiated in 2019 and their associated large disbursements materializing over the medium-term. In 2019 when the government came into power many of the projects undertaken by the former administration had been completed or were on halt, therefore capital spending was relatively low in 2019 compared to recent years. In 2019 the projects initiated by the government were in the planning and preparatory stages, with the major disbursements on these projects happening in 2020. The authorities note that the Sovereign Development Fund (SDF) established in 2017 to set aside funds for future debt repayment obligations of the Maldives, has reached a fund size of USD 223 million as of July 2020. With the recent increase in Airport Development Fee rates and the forecasted recovery in the tourist arrivals, the SDF size will continue to grow in the coming years. Currently the government is working on drafting an act to strengthen the legal status and governance of the SDF.

Figure 1. Maldives: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2021–2031



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	n.a.	n.a.
Market financing	No	No

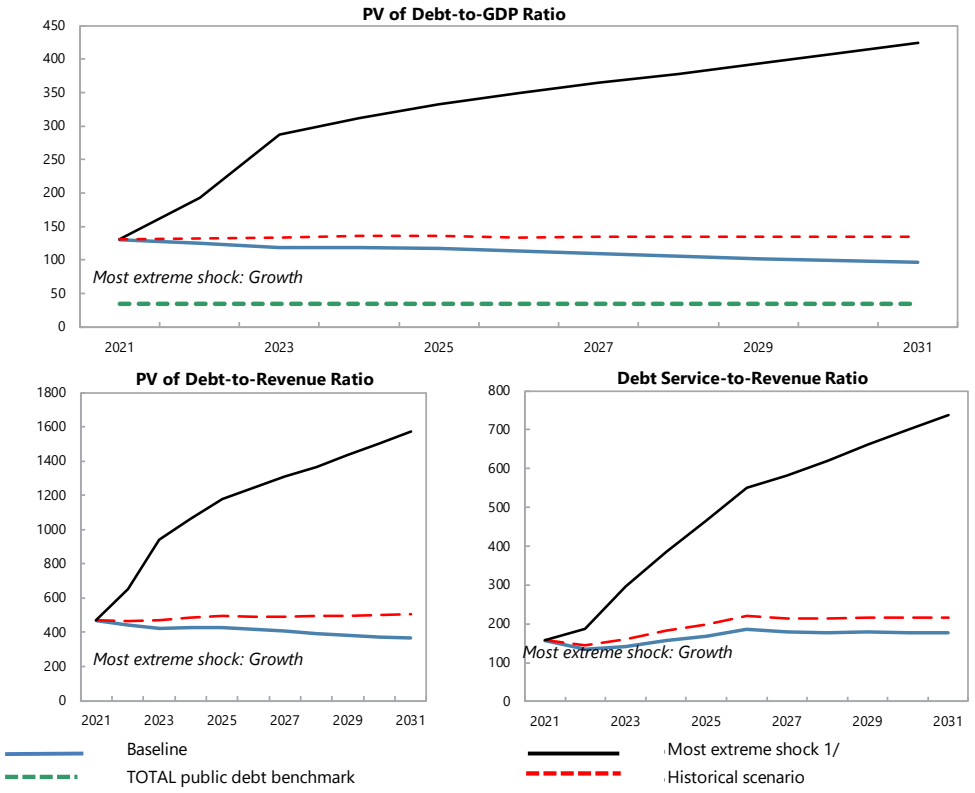
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	4	4

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Maldives: Indicators of Public Debt Under Alternatives Scenarios, 2021–2031



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	11%	11%
Domestic medium and long-term	31%	31%
Domestic short-term	58%	58%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.5%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	2.9%	2.9%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	1.9%	1.9%

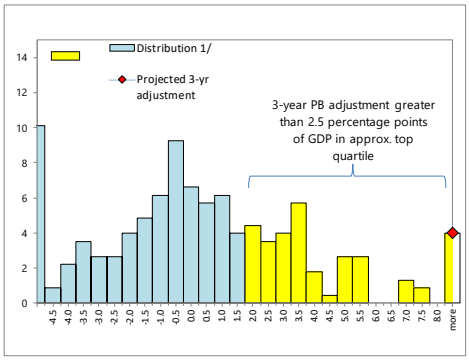
* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2031. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

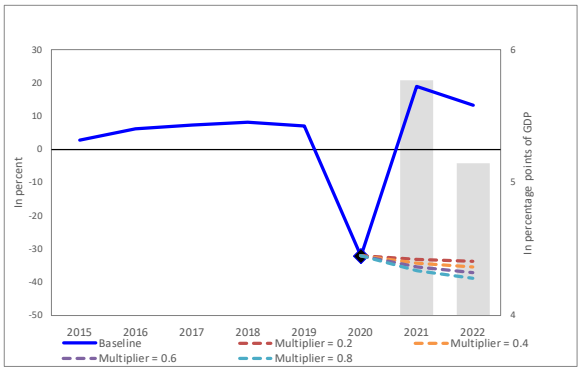
Figure 3. Maldives: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



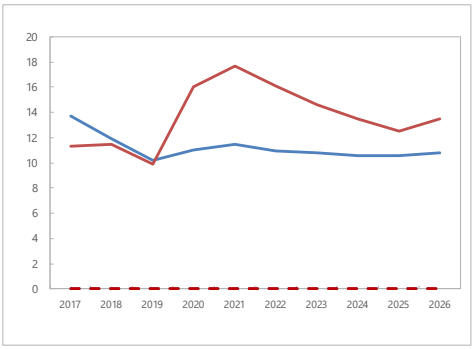
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



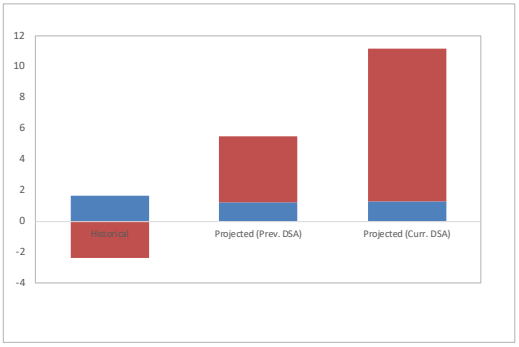
1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Gov. Invest. - Prev. DSA
Gov. Invest. - Curr. DSA
Priv. Invest. - Prev. DSA
Priv. Invest. - Curr. DSA

Contribution to Real GDP growth
(percent, 5-year average)

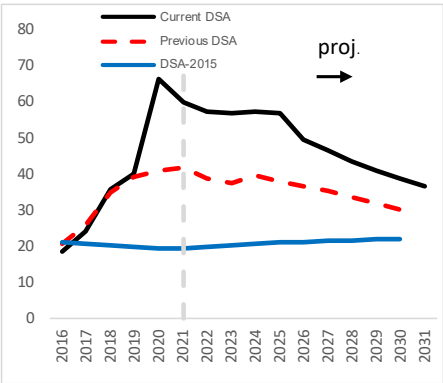


Contribution of other factors
Contribution of government capital

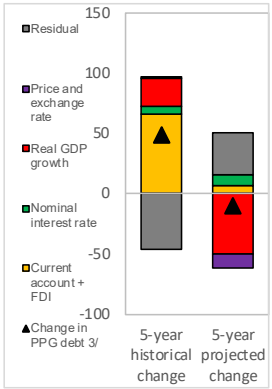
Figure 4. Maldives: Drivers of Debt Dynamics – Baseline Scenario

External debt

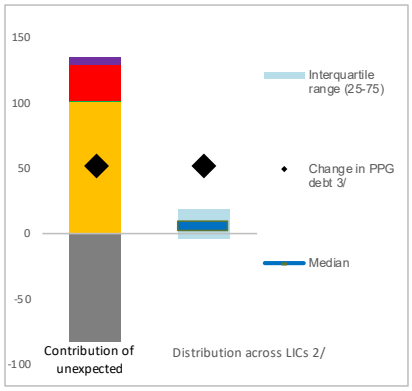
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

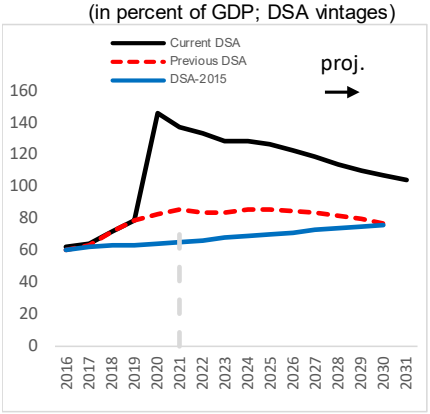


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

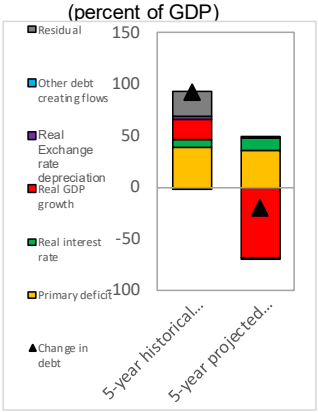


Public debt

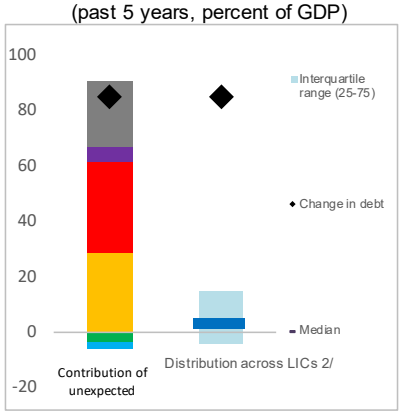
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



1/ Difference between anticipated and actual contributions on debt ratios.

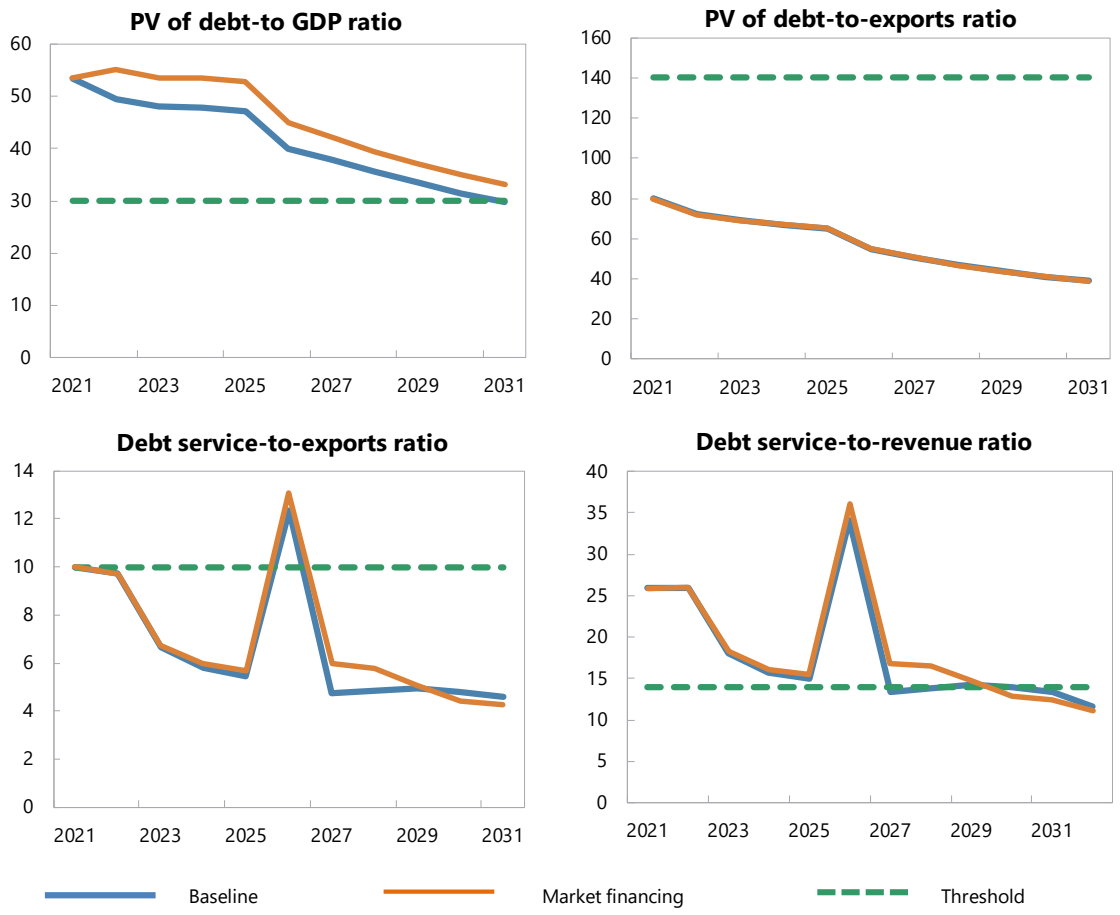
2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

Figure 5. Maldives: Market Financing Risk Indicators

	GFN 1/	EMBI 2/
Benchmarks	14	570
Values	58	n.a.
Breach of benchmark	Yes	n.a.
Potential heightened liquidity needs	Moderate	

1/ Maximum gross financing needs (GFN) over 3-year baseline projection horizon.
2/ EMBI spreads correspond to the latest available data.



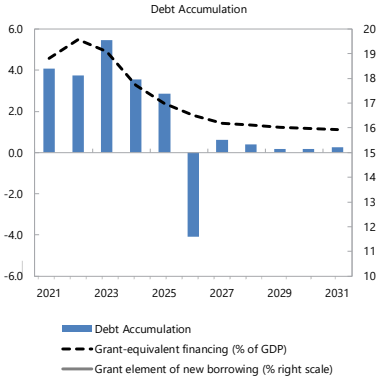
Sources: Country authorities; and staff estimates and projections.

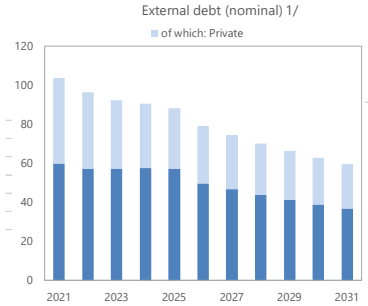
Table 1. Maldives: External Debt Sustainability Framework, Baseline Scenario, 2018–2041

(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 8/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
External debt (nominal) 1/	**62.2**	**68.9**	**119.3**	**103.5**	**96.1**	**92.1**	**90.4**	**88.2**	**79.0**	**59.4**	**41.5**	**46.4**	**80.2**
of which: public and publicly guaranteed (PPG)	**36.0**	**40.1**	**66.5**	**59.8**	**57.2**	**57.1**	**57.4**	**57.0**	**49.5**	**36.7**	**27.4**	**29.5**	**49.6**
Change in external debt	26.6	6.7	50.4	-15.8	-7.4	-4.1	-1.6	-2.3	-9.2	-3.2	-0.9		
Identified net debt-creating flows	**13.8**	**5.6**	**55.5**	**-10.4**	**-5.9**	**-9.2**	**-4.2**	**-4.0**	**-3.8**	**-1.8**	**-0.6**	**6.8**	**-5.3**
Non-interest current account deficit	**27.6**	**25.5**	**27.0**	**13.7**	**11.9**	**7.1**	**7.2**	**6.9**	**6.6**	**5.0**	**5.7**	**16.0**	**6.8**
Deficit in balance of goods and services	9.8	6.2	17.4	2.9	-2.2	-7.4	-7.1	-7.7	-7.9	-9.4	-5.5	**-1.8**	**-7.3**
Exports	67.7	66.4	47.6	66.9	68.5	69.7	71.6	72.5	73.1	76.5	77.1		
Imports	77.5	72.6	65.0	69.7	66.2	62.3	64.5	64.8	65.2	67.1	71.5		
Net current transfers (negative = inflow)	9.3	10.3	5.0	3.4	7.3	7.7	7.4	7.7	7.6	7.6	4.9	**9.3**	**7.1**
of which: official	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other current account flows (negative = net inflow)	8.5	9.0	4.7	7.5	6.8	6.8	6.9	6.9	6.9	6.8	6.3	**8.5**	**6.9**
Net FDI (negative = inflow)	**-10.9**	**-17.1**	**-9.3**	**-7.9**	**-8.0**	**-8.0**	**-8.0**	**-8.0**	**-7.5**	**-5.0**	**-5.0**	**-10.9**	**-7.0**
Endogenous debt dynamics 2/	**-2.9**	**-2.8**	**37.8**	**-16.2**	**-9.8**	**-8.3**	**-3.4**	**-2.9**	**-2.9**	**-1.8**	**-1.3**		
Contribution from nominal interest rate	0.8	0.9	2.9	2.2	2.0	1.8	1.8	1.7	1.5	0.8	0.6		
Contribution from real GDP growth	-2.6	-4.1	33.2	-18.4	-11.8	-10.1	-5.2	-4.6	-4.4	-2.6	-1.9		
Contribution from price and exchange rate changes	-1.1	0.3	1.7	...	...	...	...	...	...	...	...		
Residual 3/	**12.8**	**1.0**	**-5.1**	**-5.3**	**-1.5**	**5.2**	**2.6**	**1.7**	**-5.4**	**-1.4**	**-0.3**	**1.0**	**-0.2**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**61.4**	**53.5**	**49.4**	**48.0**	**47.8**	**47.1**	**40.0**	**29.8**	**22.5**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**128.9**	**80.0**	**72.2**	**68.9**	**66.7**	**64.9**	**54.7**	**38.9**	**29.2**		
PPG debt service-to-exports ratio	**2.4**	**5.1**	**17.7**	**10.0**	**9.7**	**6.7**	**5.8**	**5.5**	**12.4**	**4.6**	**3.3**		
PPG debt service-to-revenue ratio	**6.1**	**13.4**	**34.1**	**25.9**	**26.0**	**18.1**	**15.7**	**15.0**	**34.1**	**13.4**	**9.6**		
Gross external financing need (Million of U.S. dollars)	1242.8	904.6	1239.5	848.3	840.2	518.8	516.6	504.3	921.3	701.4	1024.7		
Key macroeconomic assumptions													
Real GDP growth (in percent)	8.1	7.0	-32.0	18.9	13.2	12.1	6.1	5.4	5.4	4.4	4.7	**2.5**	**7.8**
GDP deflator in US dollar terms (change in percent)	3.1	-0.5	-2.4	2.9	2.3	1.9	1.8	2.0	2.0	2.0	0.7	**2.1**	**2.1**
Effective interest rate (percent) 4/	2.4	1.6	2.8	2.2	2.3	2.2	2.1	2.0	1.8	1.4	1.6	**1.7**	**1.8**
Growth of exports of G&S (US dollar terms, in percent)	8.0	4.4	-52.4	71.8	18.6	16.4	10.9	8.9	8.3	6.5	6.6	**1.4**	**15.9**
Growth of imports of G&S (US dollar terms, in percent)	17.1	-0.3	-40.6	31.3	10.0	7.5	11.8	8.0	8.1	6.2	8.3	**5.8**	**10.5**
Grant element of new public sector borrowing (in percent)	...	...	...	24.4	26.8	26.5	26.2	26.0	25.6	23.7	21.5	**...**	**25.3**
Government revenues (excluding grants, in percent of GDP)	26.2	25.4	24.8	25.8	25.7	25.7	26.4	26.4	26.4	26.2	26.3	**25.2**	**26.2**
Aid flows (in Million of US dollars) 5/	53.2	74.9	92.0	193.0	264.7	270.0	193.6	150.7	124.9	108.6	131.9		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	4.6	5.5	4.9	3.3	2.4	1.8	1.1	0.7	**...**	**2.6**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	37.9	42.3	40.6	38.8	35.4	37.8	32.7	24.1	**...**	**37.3**
Nominal GDP (Million of US dollars)	5,293	5,632	3,738	4,573	5,296	6,049	6,533	7,027	7,553	10,609	19,865		
Nominal dollar GDP growth	11.5	6.4	-33.6	22.4	15.8	14.2	8.0	7.6	7.5	6.5	5.5	**4.8**	**10.1**
Memorandum items:													
PV of external debt 7/	...	...	114.1	97.2	88.3	83.1	80.8	78.2	69.5	52.4	36.5		
In percent of exports	...	...	239.7	145.5	129.1	119.1	112.8	107.9	95.0	68.5	47.4		
Total external debt service-to-exports ratio	9.9	11.4	32.5	19.1	17.5	13.6	12.2	11.4	17.9	8.7	5.8		
PV of PPG external debt (in Million of US dollars)			2294.3	2446.1	2617.4	2906.6	3121.3	3308.9	3022.1	3159.4	4463.3		
(PVt-PVt-1)/GDPt-1 (in percent)				4.1	3.7	5.5	3.5	2.9	-4.1	0.3	1.0		
Non-interest current account deficit that stabilizes debt ratio	1.0	18.9	-23.4	29.5	19.2	11.2	8.8	9.1	15.8	8.1	6.6		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

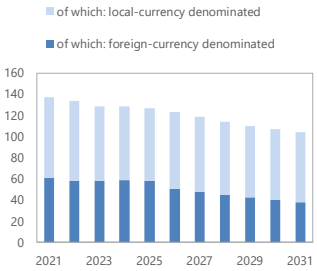
Table 2. Maldives: Public Sector Debt Sustainability Framework, Baseline Scenario, 2018–2041

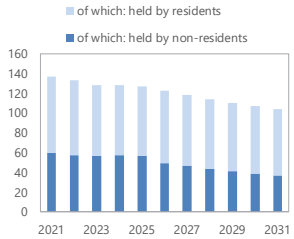
(In percent of GDP, unless otherwise indicated)

	Actual			Projections								Average 6/	
	2018	2019	2020	2021	2022	2023	2024	2025	2026	2031	2041	Historical	Projections
Public sector debt 1/	**72.0**	**78.3**	**146.0**	**137.2**	**133.3**	**128.2**	**128.4**	**126.8**	**122.9**	**104.0**	**89.8**	**70.0**	**120.9**
of which: external debt	**36.0**	**40.1**	**66.5**	**59.8**	**57.2**	**57.1**	**57.4**	**57.0**	**49.5**	**36.7**	**27.4**	**29.5**	**49.6**
Change in public sector debt	7.5	6.3	67.7	-8.8	-3.9	-5.2	0.2	-1.6	-3.8	-2.9	0.0		
Identified debt-creating flows	**-0.7**	**2.5**	**60.1**	**-6.4**	**-4.3**	**-6.0**	**-1.1**	**-1.1**	**-3.4**	**-2.5**	**-0.3**	**6.1**	**-3.5**
Primary deficit	**3.5**	**4.9**	**20.0**	**14.2**	**9.0**	**5.7**	**3.9**	**3.3**	**4.0**	**0.4**	**0.6**	**5.3**	**3.8**
Revenue and grants	27.2	26.8	27.2	27.9	28.4	28.0	27.9	27.3	27.2	26.6	26.4	**26.1**	**27.4**
of which: grants	1.0	1.3	2.5	2.1	2.7	2.3	1.4	0.9	0.8	0.4	0.1		
Primary (noninterest) expenditure	30.7	31.6	47.2	42.1	37.5	33.8	31.8	30.6	31.2	27.0	27.0	**31.5**	**31.2**
Automatic debt dynamics	**-4.9**	**-2.7**	**43.5**	**-20.6**	**-13.4**	**-11.8**	**-5.0**	**-4.5**	**-4.5**	**-2.9**	**-0.9**		
Contribution from interest rate/growth differential	-4.6	-3.6	40.7	-20.6	-13.4	-11.8	-5.0	-4.5	-4.5	-2.9	-0.9		
of which: contribution from average real interest rate	0.3	1.1	3.9	2.6	2.7	2.6	2.4	2.1	2.0	1.7	3.2		
of which: contribution from real GDP growth	-4.9	-4.7	36.8	-23.2	-16.0	-14.3	-7.4	-6.6	-6.5	-4.5	-4.1		
Contribution from real exchange rate depreciation	-0.4	0.9	2.8	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.7**	**0.3**	**-3.4**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**-2.9**	**0.0**	**0.0**	**-0.3**	**-0.3**
Privatization receipts (negative)	0.7	0.3	-3.4	0.0	0.0	0.0	0.0	0.0	-2.9	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**8.2**	**3.8**	**7.6**	**-2.4**	**0.4**	**0.9**	**1.2**	**-0.4**	**-0.4**	**-0.4**	**0.4**	**3.2**	**-0.3**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**140.9**	**130.9**	**125.5**	**119.2**	**118.7**	**116.9**	**113.4**	**97.0**	**84.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**517.2**	**468.9**	**441.9**	**425.0**	**426.1**	**428.3**	**416.8**	**364.8**	**320.9**		
Debt service-to-revenue and grants ratio 3/	**74.1**	**26.4**	**47.3**	**157.7**	**135.2**	**141.0**	**156.7**	**167.9**	**185.7**	**177.3**	**80.1**		
Gross financing need 4/	24.4	12.3	29.5	58.2	47.5	45.3	47.6	49.2	51.6	47.5	21.8		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	8.1	7.0	-32.0	18.9	13.2	12.1	6.1	5.4	5.4	4.4	4.7	**2.5**	**7.8**
Average nominal interest rate on external debt (in percent)	3.5	2.7	4.8	4.0	3.9	3.6	3.4	3.2	2.8	2.3	2.5	**2.6**	**3.0**
Average real interest rate on domestic debt (in percent)	-0.4	2.7	4.1	1.9	2.2	2.6	2.5	2.2	2.4	2.4	5.2	**0.0**	**2.4**
Real exchange rate depreciation (in percent, + indicates depreciation)	-1.6	2.7	4.6	...	...	...	...	...	...	...	...	**-0.3**	**...**
Inflation rate (GDP deflator, in percent)	3.1	-0.5	-2.4	2.9	2.3	1.9	1.8	2.0	2.0	2.0	0.7	**4.0**	**2.1**
Growth of real primary spending (deflated by GDP deflator, in percent)	13.7	10.1	1.5	6.1	0.7	1.0	-0.1	1.6	7.3	3.7	5.9	**6.7**	**2.4**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	-4.0	-1.4	-47.8	23.0	12.9	10.9	3.8	4.9	7.8	3.3	0.5	**-17.7**	**7.6**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, government-guaranteed debt . Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Maldives: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2021–2031

(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of debt-to GDP ratio											
Baseline	**53**	**49**	**48**	**48**	**47**	**40**	**38**	**36**	**33**	**31**	30
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**53**	**55**	**62**	**68**	**73**	**71**	**75**	**79**	**83**	**85**	**86**
B. Bound Tests											
B1. Real GDP growth	**53**	**72**	**99**	**99**	**97**	**83**	**78**	**73**	**69**	**65**	**62**
B2. Primary balance	**53**	**50**	**50**	**50**	**50**	**43**	**41**	**39**	**37**	**35**	**33**
B3. Exports	**53**	**73**	**109**	**107**	**104**	**95**	**90**	**83**	**76**	**71**	**66**
B4. Other flows 3/	**53**	**55**	**59**	**58**	**57**	**50**	**47**	**44**	**41**	**38**	**36**
B5. Depreciation	**53**	**62**	**60**	**59**	**59**	**50**	**47**	**44**	**41**	**39**	**37**
B6. Combination of B1-B5	**53**	**89**	**114**	**113**	**111**	**98**	**92**	**85**	**80**	**74**	**69**
C. Tailored Tests											
C1. Combined contingent liabilities	**53**	**50**	**49**	**49**	**48**	**42**	**39**	**37**	**35**	**33**	**32**
C2. Natural disaster	**53**	**51**	**50**	**50**	**50**	**43**	**41**	**39**	**37**	**35**	**34**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**53**	**55**	**54**	**53**	**53**	**45**	**42**	**39**	**37**	**35**	**33**
Threshold	30	30	30	30	30	30	30	30	30	30	30
PV of debt-to-exports ratio											
Baseline	80	72	69	67	65	55	51	47	44	41	39
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	80	81	89	95	100	97	100	105	109	111	112
B. Bound Tests											
B1. Real GDP growth	80	72	69	67	65	55	51	47	44	41	39
B2. Primary balance	80	73	71	70	68	59	55	51	48	45	43
B3. Exports	80	**142**	**279**	**266**	**257**	**231**	**215**	**195**	**179**	**165**	**153**
B4. Other flows 3/	80	81	85	82	79	68	63	58	54	50	47
B5. Depreciation	80	72	68	66	64	54	50	46	43	41	38
B6. Combination of B1-B5	80	123	93	**149**	**145**	127	116	107	99	92	86
C. Tailored Tests											
C1. Combined contingent liabilities	80	73	70	68	67	57	53	49	46	44	42
C2. Natural disaster	80	76	73	71	70	60	56	52	49	47	45
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	80	72	69	67	65	55	51	47	44	41	39
Threshold	140	140	140	140	140	140	140	140	140	140	140
Debt service-to-exports ratio											
Baseline	10	10	7	6	5	**12**	5	5	5	5	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	10	**11**	8	7	7	**17**	7	8	9	9	10
B. Bound Tests											
B1. Real GDP growth	10	10	7	6	5	**12**	5	5	5	5	5
B2. Primary balance	10	10	7	6	6	**13**	5	5	5	5	5
B3. Exports	10	**14**	**16**	**16**	**15**	**29**	**16**	**20**	**19**	**19**	**18**
B4. Other flows 3/	10	10	7	6	6	**13**	6	6	6	6	6
B5. Depreciation	10	10	7	6	5	**12**	5	5	5	5	5
B6. Combination of B1-B5	10	**13**	**12**	**11**	**10**	**22**	**11**	**11**	**11**	**11**	**10**
C. Tailored Tests											
C1. Combined contingent liabilities	10	10	7	6	6	**12**	5	5	5	5	5
C2. Natural disaster	10	**10**	7	6	6	**13**	5	5	5	5	5
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	10	10	7	6	6	**13**	6	6	5	4	4
Threshold	10	10	10	10	10	10	10	10	10	10	10
Debt service-to-revenue ratio											
Baseline	**26**	**26**	**18**	**16**	**15**	**34**	13	14	**14**	14	13
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**26**	**29**	**22**	**20**	**20**	**46**	**20**	**23**	**26**	**27**	**28**
B. Bound Tests											
B1. Real GDP growth	**26**	**38**	**37**	**33**	**31**	**71**	**28**	**29**	**29**	**29**	**28**
B2. Primary balance	**26**	**26**	**18**	**16**	**15**	**35**	14	**15**	**15**	**15**	**14**
B3. Exports	**26**	**29**	**24**	**24**	**23**	**46**	**25**	**32**	**32**	**31**	**29**
B4. Other flows 3/	**26**	**26**	**19**	**17**	**16**	**35**	**16**	**17**	**17**	**17**	**16**
B5. Depreciation	**26**	**33**	**23**	**20**	**19**	**43**	**17**	**17**	**18**	**17**	**17**
B6. Combination of B1-B5	**26**	**36**	**35**	**31**	**30**	**63**	**32**	**33**	**34**	**33**	**31**
C. Tailored Tests											
C1. Combined contingent liabilities	**26**	**26**	**18**	**16**	**15**	**34**	14	**14**	**14**	**14**	14
C2. Natural disaster	**26**	**26**	**18**	**16**	**15**	**34**	14	**14**	**15**	**14**	14
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**26**	**26**	**18**	**16**	**16**	**36**	**17**	**17**	**15**	13	12
Threshold	14	14	14	14	14	14	14	14	14	14	14

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Maldives: Sensitivity Analysis for Key Indicators of Public Debt, 2021–2031

(In percent)

	Projections 1/										
	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
PV of Debt-to-GDP Ratio											
Baseline	**131**	**126**	**119**	**119**	**117**	**113**	**110**	**106**	**102**	**100**	**97**
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	**131**	**133**	**134**	**136**	**136**	**134**	**134**	**134**	**134**	**135**	**135**
B. Bound Tests											
B1. Real GDP growth	**131**	**193**	**287**	**312**	**333**	**350**	**365**	**378**	**393**	**409**	**425**
B2. Primary balance	**131**	**131**	**130**	**129**	**127**	**123**	**119**	**115**	**111**	**108**	**105**
B3. Exports	**131**	**143**	**163**	**161**	**158**	**154**	**148**	**140**	**134**	**128**	**123**
B4. Other flows 3/	**131**	**131**	**130**	**129**	**127**	**123**	**119**	**114**	**110**	**107**	**103**
B5. Depreciation	**131**	**132**	**122**	**119**	**115**	**109**	**104**	**98**	**94**	**89**	**86**
B6. Combination of B1-B5	**131**	**143**	**159**	**167**	**173**	**177**	**179**	**181**	**184**	**187**	**191**
C. Tailored Tests											
C1. Combined contingent liabilities	**131**	**132**	**125**	**125**	**123**	**119**	**115**	**111**	**107**	**104**	**102**
C2. Natural disaster	**131**	**137**	**130**	**130**	**128**	**124**	**120**	**116**	**113**	**110**	**108**
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	**131**	**126**	**119**	**119**	**117**	**114**	**110**	**106**	**102**	**99**	**97**
TOTAL public debt benchmark	35	35	35	35	35	35	35	35	35	35	35
PV of Debt-to-Revenue Ratio											
Baseline	469	442	425	426	428	417	405	391	381	372	365
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	469	464	472	484	496	488	493	494	498	501	506
B. Bound Tests											
B1. Real GDP growth	469	652	942	1,062	1,180	1,246	1,312	1,367	1,435	1,502	1,572
B2. Primary balance	469	461	464	464	466	453	440	425	414	404	396
B3. Exports	469	502	582	579	580	564	545	518	497	478	461
B4. Other flows 3/	469	463	464	464	466	454	440	422	410	398	389
B5. Depreciation	469	469	438	428	422	403	384	364	349	335	322
B6. Combination of B1-B5	469	496	552	590	628	643	657	667	681	697	714
C. Tailored Tests											
C1. Combined contingent liabilities	469	466	447	447	449	437	424	410	399	390	383
C2. Natural disaster	469	483	464	465	468	456	445	431	421	413	406
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	469	442	425	427	429	418	405	391	380	372	364
Debt Service-to-Revenue Ratio											
Baseline	158	135	141	157	168	186	179	178	178	178	177
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2021-2031 2/	158	146	161	182	198	221	214	215	217	217	216
B. Bound Tests											
B1. Real GDP growth	158	188	297	385	466	550	581	620	663	701	738
B2. Primary balance	158	135	152	178	188	205	196	194	194	192	191
B3. Exports	158	135	143	161	173	190	186	191	191	189	188
B4. Other flows 3/	158	135	142	158	169	187	181	181	182	181	180
B5. Depreciation	158	130	137	150	160	184	172	169	170	168	168
B6. Combination of B1-B5	158	150	183	206	241	280	287	301	315	326	338
C. Tailored Tests											
C1. Combined contingent liabilities	158	135	154	168	180	196	188	186	186	185	184
C2. Natural disaster	158	137	163	176	189	205	198	196	196	195	194
C3. Commodity price	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
C4. Market Financing	158	135	141	157	168	188	182	180	179	177	176

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.